Predicting YouTube Video Views Using Machine Learning: An End-to-End Regression Approach

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Abstract

This project focuses on predicting YouTube video views using Machine Learning regression techniques. The project was developed using YouTube trending video data containing information related to video category, publishing time, title, description, tags, and trending behavior. The complete workflow included data preprocessing, exploratory data analysis, feature engineering, model development, model comparison, hyperparameter tuning, feature importance analysis, statistical P-value analysis, and prediction on unseen data.

Multiple regression models including Linear Regression, Random Forest, Extra Trees, and XGBoost were trained and evaluated. Random Forest achieved the best baseline performance with an R^2 Score of 0.5706. Therefore, it was selected for hyperparameter tuning using RandomizedSearchCV. The Tuned Random Forest achieved an R^2 Score of 0.7414 on the transformed target scale and was selected as the final model. The project demonstrates a complete end-to-end Machine Learning workflow for predicting YouTube video popularity.

Keywords

YouTube, Machine Learning, Regression, Random Forest, View Prediction, Hyperparameter Tuning, Feature Engineering.

I.Introduction

YouTube is one of the largest video-sharing platforms where millions of videos receive different levels of audience engagement. The number of views received by a video is an important indicator of its popularity. However, predicting video views is challenging because popularity depends on multiple factors such as category, publishing time, title, description, tags, and trending behavior.

Machine Learning techniques can analyze historical data and identify relationships between these factors and video views. In this project, regression algorithms were used to predict YouTube video views. The project follows an end-to-end approach starting from data preprocessing and ending with final model selection and unseen data prediction.

II.Problem Statement

The number of views received by a YouTube video varies significantly and depends on multiple factors. The relationship between video metadata and views is complex and may not be linear. Therefore, the problem addressed in this project is to develop a Machine Learning regression model that can predict YouTube video views using available video-related features.

III.Research Objectives

The main objective of this project is to develop a Machine Learning model for predicting YouTube video views.

The project aims to preprocess the dataset, analyze data patterns, create useful features, train multiple regression models, compare their performance, perform hyperparameter tuning on the best model, analyze feature importance, perform statistical validation, and predict views for unseen data.

IV.Research Questions

This project investigates whether YouTube video metadata can be effectively used to predict video views. It also examines which Machine Learning regression model provides better predictive

performance and whether hyperparameter tuning can improve the selected model.

V.Literature Review / Related Work

Machine Learning techniques are commonly used for predicting online content popularity and user engagement. Linear Regression is often used as a basic regression model, while ensemble algorithms such as Random Forest and Extra Trees are capable of learning nonlinear relationships from structured data. XGBoost is another popular algorithm used for prediction problems.

Based on these approaches, this project compares multiple regression algorithms to identify a suitable model for predicting YouTube video views.

VI.Research Gap

Many prediction projects focus mainly on model training and accuracy evaluation. This project follows a more complete Machine Learning workflow by including data preprocessing, exploratory analysis, feature engineering, multiple model comparison, hyperparameter tuning, feature importance analysis, P-value analysis, and unseen data prediction.

VII.Dataset Description

The project uses a YouTube trending videos dataset containing information related to videos from multiple countries. The dataset includes features such as category ID, publication details, trending date, title, description, tags, comments status, ratings status, and video error information.

The target variable selected for prediction is the number of video views. Since YouTube videos can have highly varying view counts, the target variable showed a highly skewed distribution. Therefore, appropriate preprocessing and transformation techniques were applied before model development.

VIII.Methodology

The project followed an end-to-end Machine Learning methodology. Initially, the dataset was cleaned and prepared for analysis. Exploratory Data Analysis was performed to understand the distribution of important variables and identify outliers

New features were created from publication dates, titles, descriptions, and tags. The processed dataset was divided into training and testing data. Multiple regression models were trained and evaluated using standard regression metrics. Random Forest was identified as the best baseline model and further optimized through hyperparameter tuning. Finally, feature importance analysis, P-value analysis, and unseen data prediction were performed.

IX.System / ML Workflow

Raw Dataset → Data Cleaning → Data Preprocessing → EDA → Outlier Analysis → Feature Engineering → Target Transformation → Train-Test Split → Model Training → Model Evaluation → Model Comparison → Hyperparameter Tuning → Final Model Selection → Feature Importance Analysis → P-Value Analysis → Unseen Data Prediction

X.Data Preprocessing

Data preprocessing was performed to make the dataset suitable for Machine Learning. Missing and inconsistent values were examined, and date-related columns were converted into appropriate datetime formats.

A new feature called days_to_trending was created by calculating the difference between the publication date and trending date. The feature was validated, resulting in zero missing values and zero negative values.

The target variable, views, had a highly skewed distribution. Therefore, logarithmic transformation was applied to reduce skewness and improve model learning.

XI.Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify important patterns. Variables such as views, likes, dislikes, and comment count were analyzed.

The analysis revealed the presence of significant outliers. These outliers were not simply removed because extremely high engagement values can represent genuinely popular or viral YouTube videos. Removing them could result in the loss of meaningful real-world information.

The distribution of views was highly skewed, which supported the use of logarithmic transformation for the target variable.

XII.Feature Engineering / Feature Selection

Several new features were engineered from the available data. Date-related features were extracted from publication information. Text-related features such as title length, title word count, description length, description word count, and tag count were created.

The days_to_trending feature was also included to represent the time taken by a video to appear in trending data.

After training the final model, feature importance analysis was performed. The preprocessing pipeline generated a total of 19 processed features, and feature importance values were extracted from the final Random Forest model.

XIII. Model Development

Four regression models were developed and evaluated in this project:

- Linear Regression
- Random Forest Regressor
- Extra Trees Regressor
- XGBoost Regressor

Linear Regression was used as a basic baseline model. Ensemble-based algorithms were used because they can capture complex nonlinear relationships between the input features and the target variable.

All models were trained using the processed dataset and evaluated using the same regression metrics.

XIV. Experimental Setup

The dataset was divided into training and testing datasets. The target variable was transformed using logarithmic transformation due to its highly skewed distribution.

A preprocessing pipeline was used to ensure consistent feature transformation. Models were trained using the training data and evaluated on the test data.

The experiments used Mean Absolute Error, Root Mean Squared Error, and R^2 Score for model evaluation. RandomizedSearchCV was used for hyperparameter optimization of the best-performing model.

XV. Model Validation

The trained models were validated using test data that was not used during training. This approach helped evaluate the ability of the models to generalize to new data.

The performance of each model was measured using MAE, RMSE, and R^2 Score. The R^2 Score was mainly used to compare the predictive capability of the regression models.

XVI. Hyperparameter Tuning

Random Forest achieved the best performance among the baseline models. Therefore, it was selected for hyperparameter tuning.

RandomizedSearchCV was used instead of GridSearchCV because it efficiently searches different parameter combinations while reducing computational cost. The tuning process tested 10 parameter combinations using 3-fold cross-validation, resulting in a total of 30 model fits.

The best parameters obtained were:

- n_estimators: 200
- min_samples_split: 5
- min_samples_leaf: 1
- max_features: log2
- max_depth: 30

The best cross-validation R^2 Score obtained during tuning was **0.7032**.

XVII. Model Evaluation

The models were evaluated using three regression metrics.

Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted values.

Root Mean Squared Error (RMSE) measures prediction error while giving more importance to larger errors.

R^2 Score measures how well the model explains the variation in the target variable. A higher R^2 Score indicates better predictive performance.

For the final Tuned Random Forest model, the evaluation results were:

- MAE: **0.7085**
- RMSE: **0.9921**
- R^2 Score: **0.7414**

These MAE and RMSE values correspond to the log-transformed target scale used during model evaluation.

XVIII. Model Comparison

The baseline models were compared using MAE, RMSE, and R^2 Score.

Model	MAE	RMSE	R^2 Score
Random Forest	**668,615.2**	**4,620,841**	**0.570592**
Extra Trees	815,108.8	5,129,534	0.470844
XGBoost	883,899.4	5,177,896	0.460819
Linear Regression	1,235,420	7,120,895	-0.019757

Among the baseline models, Random Forest achieved the highest R^2 Score of **0.570592**. Therefore, Random Forest was selected for hyperparameter tuning.

After tuning, the Tuned Random Forest achieved an R^2 Score of **0.741365**, showing a clear improvement in predictive performance.

R^2 Score Improvement

Model	R^2 Score
Baseline Random Forest	0.570592
Tuned Random Forest	**0.741365**

The improvement in R^2 Score demonstrates the effectiveness of hyperparameter optimization.

XIX. Final Model Selection

The **Tuned Random Forest Regressor** was selected as the final model.

The baseline Random Forest achieved an R^2 Score of **0.570592**, while the tuned model achieved an improved R^2 Score of **0.741365**.

The tuned model also achieved a cross-validation R^2 Score of **0.703237**, indicating consistent performance during the tuning process. Based on the improved predictive performance, Tuned Random Forest was selected as the final model for feature importance analysis and unseen data prediction.

XX. Sample / Unseen Data Prediction

The final Tuned Random Forest model was used to predict values from unseen test data. The initial predictions were generated on the logarithmic scale because the target variable was log-transformed during training.

Some sample predictions were:

Actual Log Views	Predicted Log Views	Absolute Error
13.626545	13.066809	0.559736
13.824811	12.872480	0.952332
10.849298	11.400081	0.550783
7.972811	10.214360	2.241549
12.006121	11.392757	0.613363
10.329213	10.285269	0.043944

The predictions were subsequently converted back to the original view-count scale using inverse logarithmic transformation. This allowed the predicted values to be interpreted as actual YouTube view counts.

The unseen predictions demonstrate that the final model can generate predictions for data samples that were not used during model training.

XXI. Results

The project successfully developed and evaluated multiple Machine Learning regression models for YouTube video view prediction.

Among the baseline models, Random Forest performed best with an R^2 Score of **0.570592**. Hyperparameter tuning improved the Random

Forest model, resulting in a final R^2 Score of **0.741365**.

The best cross-validation R^2 Score obtained during hyperparameter tuning was **0.703237**. The improvement from the baseline model to the tuned model indicates that selecting suitable hyperparameters significantly improved predictive performance.

The final model was further analyzed using feature importance and statistical P-value analysis.

XXII. Discussion

The results indicate that ensemble-based models are more effective than simple Linear Regression for predicting YouTube video views. Linear Regression achieved a negative R^2 Score, indicating that a simple linear relationship was insufficient for modeling the dataset.

Random Forest performed best because it can capture nonlinear relationships and interactions between multiple features. Hyperparameter tuning further improved its performance.

Feature importance analysis was performed on 19 processed features to understand how different video-related characteristics contributed to prediction. Statistical analysis using OLS regression showed that **17 out of 19 features were statistically significant**, while 2 features were not significant.

The overall OLS model P-value was **0.0**, which is below the significance level of 0.05. This indicates a statistically significant overall relationship between the selected features and the target variable. However, the P-value analysis was used as supplementary statistical validation because traditional tree-based Random Forest models do not directly provide P-values.

XXIII. Limitations

The project has certain limitations. YouTube video popularity is influenced by several external factors that are not available in the dataset, such as channel subscriber count, thumbnail quality, recommendation algorithms, social media promotion, and changing audience preferences.

The dataset also contains highly variable view counts, especially for viral videos. Although logarithmic transformation helped reduce skewness, predicting extremely viral videos remains challenging.

Additionally, the statistical P-values were obtained using an OLS model and not directly from the Random Forest algorithm.

XXIV. Conclusion

This project successfully developed an end-to-end Machine Learning regression system for predicting YouTube video views using trending video data.

The project included data preprocessing, exploratory data analysis, outlier analysis, feature engineering, target transformation, model development, model comparison, hyperparameter tuning, feature importance analysis, statistical P-value analysis, and unseen data prediction.

Four regression models were evaluated: Linear Regression, Random Forest, Extra Trees, and XGBoost. Among the baseline models, Random Forest achieved the best R^2 Score of **0.570592**.

Hyperparameter tuning was performed using RandomizedSearchCV with 10 parameter combinations and 3-fold cross-validation. The Tuned Random Forest achieved a cross-validation R^2 Score of **0.703237** and a test R^2 Score of **0.741365**. Therefore, the Tuned Random Forest Regressor was selected as the final model.

The statistical analysis also showed that 17 out of 19 features were significant, and the overall model P-value was below 0.05. The project demonstrates that Machine Learning, particularly ensemble tree-based models, can effectively analyze YouTube video characteristics and predict video popularity.

Future Scope

The project can be further improved by including additional information such as channel subscriber count, channel history, thumbnail features, sentiment analysis of titles and descriptions, and social media engagement.

Future work can also explore deep learning models and real-time YouTube data. Additional feature engineering techniques and advanced ensemble methods may further improve prediction performance.

References

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Appendix

Models Used

- Linear Regression
- Random Forest Regressor
- Extra Trees Regressor
- XGBoost Regressor
- Tuned Random Forest Regressor

Final Random Forest Hyperparameters

n_estimators = 200

max_depth = 30

min_samples_split = 5

min_samples_leaf = 1

max_features = log2

random_state = 42

Final Model Performance

Metric	Value
Best Cross-Validation R^2	0.703237
Final Test R^2	**0.741365**
MAE (Log Scale)	0.708517
RMSE (Log Scale)	0.992129
Significant Features	17 / 19
Overall OLS P-value	0.0

Final Model

Tuned Random Forest Regressor